

## INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Mineral Exploration      L-T-P:3 – 0 – 0      Credits:3

3. Pre-requisites: Economic Geology

4. Syllabus and reference books:

### Syllabus:

Mineral Exploration as a high risk Economic activity, Macroeconomic considerations of Mineral Exploration; The 4 stage Architecture of Mineral Exploration – reconnaissance to exploitation conforming to UNFC classification scheme; The Science of Mineral Exploration – application of Ore genetic models as guides to systematic mineral exploration with examples; Geological methods – use of structural, stratigraphic and geomorphic guides, Indicator Minerals and their importance in exploration; Geochemical methods – primary and secondary dispersion halos, Litho, pedo and hydro and atmo geochemical methods, detection of anomalies; Geophysical methods – electrical, gravity and magnetic methods applied to exploration of different types of metallic mineral deposits; Introduction to computerized mineral exploration – mineral potential mapping using data driven methods; Introduction to fractals and multifractals and application to Mineral Exploration; Important case studies

### Reference Books:

- 1) Mineral Exploration Principles and Applications by S. K. Haldar (Elsevier)
- 2) Looking into the Earth by Mussett and Khan (Cambridge Univ Press)
- 3)

5. Lecture-wise break-up:

| Sl. No.               | Topic                                                                  | No. of lectures |
|-----------------------|------------------------------------------------------------------------|-----------------|
| 1.                    | Introductory concepts                                                  | 3               |
| 2.                    | Four stage architecture of Mineral Exploration                         | 6               |
| 3.                    | Application of Ore Genetic Models to Mineral Exploration with Examples | 12              |
| 4.                    | Geologic and Geochemical Methods                                       | 6               |
| 5                     | Geophysical Methods                                                    | 6               |
| 6                     | Advanced concepts and methods                                          | 3               |
| Total number of hours |                                                                        | 36              |